Dhruv H

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Summary

Third year Information Science student at BMS College of Engineering with interests in backend engineering, distributed systems, and machine learning. I enjoy building scalable, performance oriented software and turning theoretical concepts into practical, reliable systems through projects spanning backend infrastructure, ML research, and data-intensive applications.

Education

B.M.S College of Engineering — Bengaluru, Karnataka
Bachelor of Engineering in Information Science — *Aug. 2023 – Present*
- CGPA: 9.18/10.0

Experience

Profinch Solutions — Bengaluru, Karnataka
Software Engineering Intern — *Jul 2026 – Present*
- Built a Grafana style observability platform (FastAPI, React/TypeScript, MySQL, Docker) with 13 relational models, a connector abstraction for Oracle/PostgreSQL/MySQL sources, and a per-widget APScheduler pipeline with a 4 rule statistical anomaly engine (z-score, moving-average, above-average, below-count).
- Implemented an entity scoped dashboard system (per region widget views via query-level sentinel substitution) with admin level widget replication/override and JWT + RBAC access control down to per dashboard grants.

Projects

Exoplanet Atmospheric Retrieval | *Python, XGBoost, Scikit-learn, Optuna* | *Manuscript under review, Scientific Reports*
- Owned model training and pipeline design on a team research project benchmarking MLP, SVR, and XGBoost on the Ariel Big Data Challenge dataset to retrieve 5 gas abundances and equilibrium temperature from transmission spectra; XGBoost generalised best to real JWST NIRSpec observations of WASP-39b, matching published retrieval results.
- Identified preprocessing as the dominant performance factor (52-bin wavelength interpolation, log-scaling, sample-wise normalisation), and validated generalisation on out-of-distribution real JWST observational data across all five atmospheric gases.

QueryMind: Natural language to SQL platform | *FastAPI, PostgreSQL, Redis, Kafka, ChromaDB, Gemini, React*
- Built a text-to-SQL platform where users upload CSV/Excel/JSON files and query in plain English; two-stage RAG pipeline retrieves DDL embeddings and similar SQL examples from a curated Spider/WikiSQL pool in ChromaDB, with Gemini 2.5 Flash generating validated SQL executed against isolated per-user PostgreSQL schemas with Redis result caching and JWT auth.
- Designed event-driven architecture with Kafka (KRaft) across 4 topics and an Audit Consumer persisting all events to PostgreSQL, with Redis sliding window rate limiting, read-only SQL validation, and results rendered as paginated tables or interactive Recharts visualizations.

dvara: Malicious URL detection via Bloom Filter | *Python, FastAPI, Redis, PyPI*
- Published a PyPI package using a Bloom Filter (the same technique used by Chrome Safe Browsing) to detect malicious URLs at 1/100th the memory of a hash set, indexing 268K+ threats from URLhaus, PhishTank, OpenPhish, and CERT Polska into a 5.14MB filter sized for 3M URLs at 0.1% target FPR.
- Designed a two stage lookup (∼0.003ms bloom check, 145K URLs/sec, PostgreSQL confirmation only on hits) validated to 0 false negatives and 0 false positives across 100K test URLs, and self-hosted the full stack (FastAPI, PostgreSQL, Redis) on a single AWS EC2 instance via Docker Compose.

Technical Skills

Languages: C, C++, Python, Java, JavaScript, HTML/CSS
Frameworks/Libraries: FastAPI, Express.js, Node.js, React, Spring Boot, PyTorch, LangChain, Pandas, NumPy
Infrastructure & Cloud: Redis, PostgreSQL, MySQL, MongoDB, ChromaDB, Linux Networking, Docker, Docker Compose, Kafka, AWS
Developer Tools: Git, GitHub,Linux, UNIX, Postman, Jupyter Notebook
Concepts: Data Structures & Algorithms, Probabilistic Data Structures, Distributed Systems, OOP, REST APIs, Microservices, Multithreading, Network Programming, TCP/IP, Machine Learning, Deep Learning, Retrieval Augmented Generation (RAG), Vector Embeddings, WebSockets, Competitive Programming (Leetcode/Hackerrank)

Certifications

Nutanix Certified Associate (NCA): Hyperconverged infrastructure and Nutanix AOS administration.

JPMorgan Chase: Software Engineering Job Simulation (Forage): Kafka integrated Spring Boot microservice with JPA and REST.

NVIDIA Generative AI Explained: Fundamentals of generative AI, GANs, and transformers.